FACILITATOR GUIDE

SCENARIO 1 — S1 · Acute severe asthma in the ED

34-year-old with known asthma, escalating reliever use, arrives in severe respiratory distress. Work the case from triage to disposition.

Objectives: Classify T2 phenotype with eosinophils/FeNO · Score GINA symptom control & ACT · Deliver the evidence-based first-hour ED bundle · Escalate: continuous nebs + IV magnesium · Judge NIV honestly (Evidence D) · Discharge on GINA Step 4 MART

Scene 1: Triage: speaking in phrases, diffuse wheeze, accessory muscle use. Spirometry from clinic last month: FEV1 38% predicted during flares (illustrative). Labs today: blood eosinophils 420 cells/uL, FeNO 58 ppb (never on high-dose ICS).
✓ Best action: T2-high: eosinophils >=300 cells/uL plus FeNO >50 ppb (ATS high band)
⌁ Discuss: Your patient was on prednisone last week and the eosinophil count came back 80. A colleague concludes T2-low disease. Argue both sides — then decide when you would recheck biomarkers.

- Steroids (oral and inhaled) suppress blood eosinophils and FeNO — a single low value on treatment does not exclude T2 disease.
- Repeat biomarkers when steroid exposure is lowest; repeated measures raise diagnostic yield, and many payers accept the highest historical value.
- T2-low requires FeNO <=20 AND eos <=150 — and is itself an unstable phenotype that can declare T2 later.
- Practical rule: never phenotype a patient off a single number drawn mid-exacerbation or on OCS.

Scene 2: History (past 4 weeks): daytime symptoms daily, night waking twice weekly, reliever use daily, stopped playing soccer. ACT score 14.
✓ Best action: UNCONTROLLED — all 4 GINA criteria positive; ACT 14 = very poor control

Scene 3: Build the FIRST-HOUR ED bundle. Select every indicated intervention (and none that are wrong).
✓ Best action: Albuterol 2.5–5 mg neb q20min x3 (or pMDI+spacer 4–8 puffs q20min) + Ipratropium 0.5 mg neb q20min x3, then PRN + Prednisone 40–50 mg PO within 1 hour
⌁ Discuss: The ipratropium NNT of 5 comes from an FEV1 <50% subgroup, and the benefit disappears after admission. How should subgroup-derived, time-limited evidence change what you actually do at the bedside?

- Benefit concentrates in the sickest patients and in the ED window — front-load ipratropium early rather than continuing it on the ward.
- Subgroup findings are hypothesis-strengthening, not ironclad — but a cheap, safe drug with NNT 5 in the right subgroup shifts the threshold to act.
- Contrast with a subgroup signal that did NOT survive testing: nebulized magnesium (3Mg trial).

Scene 4: Partial response after the first hour (illustrative FEV1 now ~28% predicted). HR 124, K+ 3.1, lactate 3.4 on continuous nebs.
✓ Best action: Switch to CONTINUOUS neb albuterol 10–15 mg/hr + IV magnesium sulfate 2 g over 20 min (single dose)

Scene 5: A colleague suggests: "Just put her on BiPAP and give a little midazolam so she tolerates the mask."
✓ Best action: NIV not required — she is improving. If trialed: IPAP 8/EPAP 4 (up to ~12/5), close monitoring, and NEVER sedate for mask tolerance
⌁ Discuss: A colleague argues the observational data (intubation RR ~0.46–0.55) justify routine NIV in severe asthma. What is wrong — and what is right — about that argument?

- Confounding by indication: patients selected for NIV are systematically different from those intubated first.

> - The RCTs are small and used modest pressures (IPAP 8–12 / EPAP 4–5); GINA therefore gives no formal recommendation (Evidence D).
> - What IS defensible: a closely monitored trial in a cooperative patient with an escape plan — never sedation for tolerance, never in agitation or obtundation.

> **Scene 6:** 6 hours later: comfortable, wheeze-free, illustrative FEV1 68% predicted. Build the DISCHARGE package.
> ✓ Best action: Start GINA Step 4 MART: budesonide-formoterol 160/4.5 — 2 inh BID + 1 inh PRN (max 12 inh/day) + Prednisolone 40–50 mg daily x5–7 days, no taper + Follow-up within 2–7 days
> ⌁ Discuss: This patient arrived on SABA-only rescue despite a prior ED visit. Walk the system failures between that visit and today — where does the chain usually break in your shop?
> - Discharge without a controller step-up is the most fixable failure: MART cuts severe exacerbations ~32% vs same-dose ICS-LABA.
> - SABA-only rescue reinforces reliever culture and masks worsening inflammation.
> - The 2–7 day follow-up IS the intervention: technique, adherence, action plan, step-down planning at 2–3 months.
> - Ask the room: who owns that follow-up appointment before the patient leaves your ED?

Debrief — key principles:

- T2-high = eos >=300 cells/uL or FeNO >50 ppb (ATS bands: <25 low, 25–50 intermediate, >50 high); a single low eos does not exclude T2. [GINA 2026; Khatri ATS 2021]
- First hour: albuterol 2.5–5 mg neb q20min x3 + ipratropium 0.5 mg q20min x3 (RR 0.51, NNT 5 at FEV1 <50%) + prednisone 40–50 mg PO (oral = IV, Evidence A). [GINA 2026; Rodrigo AJRCCM 2000; Lazarus NEJM 2010]
- Failure of initial Rx / FEV1 <25–30% → continuous nebs 10–15 mg/hr + IV MgSO4 2 g over 20 min; nebulized Mg is ineffective. [Cochrane Camargo 2003; Kew Cochrane 2014; 3Mg Lancet 2013]
- SABA toxicity (tachycardia, hypokalemia, lactic acidosis) mimics worsening asthma — treat the patient, not the lactate. [GINA 2026]
- NIV: Evidence D, no GINA recommendation; never in agitation, never sedate for tolerance. [GINA 2026; Soroksky Chest 2003]
- Discharge = GINA Step 4 MART budesonide-formoterol 160/4.5 (2 BID + 1 PRN, max 12/day) + prednisolone 40–50 mg x5–7 d + review in 2–7 days. [GINA 2026]

SCENARIO 2 — S2 · Beyond first-line therapy

26-year-old, prior ICU admissions, 2 days of worsening despite home therapy — and the first hour in the ED has not gone the way anyone hoped. How far can you go, and when do you change course?

Objectives: Read the "normalizing PaCO2" danger sign · Deploy the rescue ladder with honest evidence grades · Choose intubation on clinical triggers, not one number · Perform RSI: ketamine + BiPAP preoxygenation · Avoid post-intubation pitfalls (sustained paralysis)

> **Scene 1:** One hour of maximal first-line therapy. He is word-dyspneic, diaphoretic, RR 32.
> ✓ Best action: Ominous: a NORMAL PaCO2 in a distressed asthmatic = fatigue; the rising-CO2 trajectory signals impending failure
> ⌁ Discuss: Teach your intern: why do we read this 'normal' gas as an emergency? Build the general rule for interpreting blood gases against expected compensation.
>
> - A distressed asthmatic should be hypocapnic; PaCO2 41 means minute ventilation is failing toward fatigue.
> - Always ask what the gas SHOULD be for this degree of distress; trajectory beats any single value.
> - Pair every gas with a work-of-breathing exam — the gas lags the muscles.

> **Scene 2:** Climb the rescue ladder. IV magnesium was already given. Select what you would ADD now (evidence grades vary — the Evidence popovers keep you honest).
> ✓ Best action: Heliox 70:30 (or 80:20) driven nebs — trial in the non-responder

> **Scene 3:** RT asks about a BiPAP trial while rescue therapy runs. He is exhausted but cooperative, protecting his airway.
> ✓ Best action: Short, closely-monitored trial: IPAP ~12 / EPAP 5, intubation kit open at the bedside, abort for agitation or any deterioration

> **Scene 4:** 30 min later: he is drowsy, mumbling, chest nearly SILENT. SpO2 88% on NIV. Repeat gas: pH 7.24 / PaCO2 68 / PaO2 58.
> ✓ Best action: INTUBATE NOW — but semi-electively: controlled conditions, volume-load first, most experienced operator
> ⌁ Discuss: Your attending pushes back: 'He is 26. The NIV cohorts show lower mortality. Why not one more hour?' Defend your timing out loud.
>
> - Strong triggers end the debate: altered mental status, silent chest, exhaustion — NIV in an obtunded patient is an aspiration event in progress.
> - Semi-elective NOW beats a crash tube in 30 minutes: ~45% complication rate demands controlled conditions and volume-loading.
> - Name the roles before you push drugs: who intubates, who watches the pressures, who owns the vent settings.

> **Scene 5:** Run the RSI. Select every element of the evidence-based approach.
> ✓ Best action: Ketamine 1–2 mg/kg IV for induction + Preoxygenate WITH the BiPAP he is already on + Post-intubation sedation: ketamine infusion 0.5–2 mg/kg/h (or propofol)
> ⌁ Discuss: Ketamine: beloved induction agent, failed bronchodilator. Why do the RCTs and bedside impressions diverge so often?
>
> - As induction: bronchodilatory + hemodynamically stable — the NEAR registry favorite (51.8%). That claim is solid.
> - As therapy: RCTs negative; bedside 'responses' ride on co-interventions and regression to the mean.
> - General lesson: separate pharmacologic plausibility from proof that a drug changes outcomes.

Debrief — key principles:

- A normalizing/rising PaCO2 in a distressed asthmatic is a danger sign; PaO2 <60 + PaCO2 >=45 = respiratory failure. Trajectory > absolute value. [GINA 2026; Papiris Crit Care 2002]
- Intubation triggers: altered mental status, silent chest, exhaustion (most common), hemodynamic instability, arrest, refractory hypoxemia. No single PaCO2 mandates it. [GINA 2026]
- Once decided: SEMI-ELECTIVE intubation — volume-load first; expect peri-intubation hypotension from hyperinflation (~45% complication rate). [Zimmerman CCM 1993; Sydow Minerva 2003]
- RSI: ketamine 1–2 mg/kg IV (NEAR 51.8%), BiPAP preoxygenation (62.9%), first-pass ~90.8%; then ketamine 0.5–2 mg/kg/h or propofol; avoid sustained paralysis (steroid myopathy). [Godwin Respir Care 2020; Garner Chest 2022]
- Rescue ladder honesty: heliox helps the sickest (Evidence B, not routine); terbutaline SC 0.25 mg / epi 0.3–0.5 mg IM are options; ketamine-as-bronchodilator has negative RCTs; inhalational anesthetics are case-series rescue for the ventilated. [Rodrigo AAAI 2014; FDA terbutaline 2022; La Via EJCP 2022; Ho Crit Care 2024]

SCENARIO 3 — S3 · Post-intubation crisis

The airway is finally secured after a difficult intubation and prolonged bagging — and the alarms are just getting started.

Objectives: Interpret absent lung sliding in context · Use the lung point to separate PTX from a non-ventilated lung · Verify ETT depth: 20/22 cm rule, tip 2–6 cm above carina · Correct a mainstem intubation and re-verify

> **Scene 1:** Intubation was difficult: 2 attempts, prolonged BVM. Now: PIP 58 cmH2O, SpO2 88% and drifting, absent LEFT breath sounds.
> ✓ Best action: Absent sliding + barcode but NO lung point after a difficult intubation → suspect RIGHT MAINSTEM (whole left lung not ventilated), not pneumothorax
> ⌁ Discuss: You fought a difficult airway and now the sats are falling — why is pneumothorax the diagnosis your brain reaches for first, and what forced you off it here?
>
> - Availability and anchoring bias: barotrauma is the feared complication, so absent sliding 'confirms' it — unless you demand the lung point.
> - A systematic post-intubation scan (tube depth, bilateral sliding, waveforms) beats pattern-matching under adrenaline.
> - Cost of the wrong call: needling a mainstem creates the pneumothorax you feared.

> **Scene 2:** Stat supine CXR (left). The ETT was secured at 26 cm at the incisors in this 168-cm woman.
> ✓ Best action: ETT tip BELOW the carina in the right mainstem with left-lung volume loss — depth should be ~20 cm (women) / 22 cm (men), tip 2–6 cm above carina

> **Scene 3:** Fix it. Select the complete correction sequence.
> ✓ Best action: Withdraw the ETT to ~21 cm under direct observation + Re-verify: bilateral lung sliding on US + repeat CXR (tip 2–6 cm above carina)
> ⌁ Discuss: Auscultation missed 55% of mainstem intubations. What other bedside 'reassurances' in your ICU deserve the same skepticism — and what verification culture replaces them?
>
> - Equal breath sounds, tube misting, symmetric chest rise: all fail at meaningful rates.
> - Replace with: the depth rule (20/22 cm), bilateral sliding on US (sens 93%/spec 96%), CXR tip 2–6 cm above carina.
> - Invite the room: name one verification step your unit skips when it is busy.

Debrief — key principles:

- Absent lung sliding is a finding, not a diagnosis: mainstem intubation silences the whole contralateral lung. The LUNG POINT separates PTX (present; spec 99–100%) from a non-ventilated lung (absent). [Patel JOMI EFAST 2021; SCCM CCM 2015]
- One B-line excludes PTX at that interspace. [Buda NEJM 2023]
- Auscultation missed 55% of mainstems; ETT depth is the best single test (88%): 20 cm women / 22 cm men; radiographic tip 2–6 cm above carina; US sens 93%/spec 96%. [Sitzwohl BMJ 2010; Ramsingh Anesthesiology 2016]
- Fix = withdraw the ETT and re-confirm with bilateral sliding + CXR — do not exchange a working tube in a difficult airway. [Schwartz CCM 1994]
- Ultrasound beats supine CXR for PTX: sens ~0.83–0.95 vs ~0.28–0.52. [SCCM CCM 2015; Sheng Respiration 2023]

SCENARIO 4 — S4 · Sudden deterioration on the ventilator

Two hours of hard-won stability on the ventilator. Then everything happens at once.

Objectives: Recognize tension PTX from ventilator signals + hemodynamics · Treat before imaging — clinical diagnosis · Apply US criteria incl. lung point; deep sulcus on supine CXR · Choose decompression site and chest tube size for a ventilated patient

> **Scene 1:** Sudden alarms: PIP 48 → 70 cmH2O over minutes, Pplat rising, SpO2 92 → 84%, then BP 78/50, HR 138.
> ✓ Best action: Suspect TENSION PNEUMOTHORAX — this is a clinical diagnosis; prepare to treat BEFORE imaging while you scan
> ⌁ Discuss: Tension pneumothorax is a clinical diagnosis — yet we all still reach for imaging. Where exactly is the line between confirming and dangerously delaying?
>
> - The numbers set the line: 70% of ventilated tension PTX deteriorate within minutes; hypotension odds 12.6x, arrest 17.7x.
> - Bedside US while the kit opens = confirmation without delay; CT = delay; in arrest, decompress empirically.
> - Agree as a team in advance which findings trigger the needle without waiting for any image.

> **Scene 2:** Rapid RIGHT chest ultrasound while the kit opens (image at left): sliding ABSENT, no B-lines, barcode on M-mode — and this time a clear transition point where sliding lung meets still lung.
> ✓ Best action: Absent sliding + absent B-lines + barcode + LUNG POINT = pneumothorax, pathognomonic

> **Scene 3:** BP now 68/42. Choose your immediate decompression.
> ✓ Best action: 14 G needle, 4th/5th ICS anterior/mid-axillary line, over the top of the rib (predicted failure ~13%)
> ⌁ Discuss: Your hospital stocks 5-cm 14G catheters and this patient has a BMI of 44. Walk through site selection and every failure mode you can name.
>
> - Chest wall depth drives failure: 2nd ICS MCL fails ~38% with a 5-cm catheter — yet is preferred in the obese (thinner wall there).
> - 4th/5th ICS anterior/mid-axillary fails ~13% with better landmark accuracy in most adults.
> - Failure modes: kinking, clot, catheter too short, wrong interspace, below-rib neurovascular injury — finger thoracostomy bypasses them all.

> **Scene 4:** Rush of air; BP recovers to 96/60. Definitive management for this VENTILATED patient with an ongoing air leak?
> ✓ Best action: Chest tube 24–28 Fr, 4th–6th ICS in the safe triangle

Debrief — key principles:

- Ventilated tension PTX: rising PIP (VC) / falling Vt (PC), rising Pplat, EARLY hypoxemia (91.8%), then hypotension (odds 12.6x) and arrest (17.7x); 70% deteriorate within minutes. Clinical diagnosis — treat before imaging. [Roberts Ann Surg 2015; Barton JEM 1997]
- US criteria: absent sliding + absent B-lines + barcode + LUNG POINT (spec 99–100%; absent in massive PTX). US sens ~0.83–0.95 vs supine CXR ~0.28–0.52. [SCCM CCM 2015; Sheng Respiration 2023]
- Supine CXR signs: deep sulcus, basal hyperlucency, double-diaphragm, mediastinal shift — sens only ~13–53%. [Gordon Radiology 1980; Chiles CCM 1986]
- Needle decompression: 14 G over the rib top; 4th/5th ICS anterior/mid-axillary (~13% failure) vs 2nd ICS MCL (~38%, but preferred in obese). Finger thoracostomy is a robust alternative. [WSES-AAST 2025; Laan Injury 2016; Inaba JTACS 2015]

- Definitive: chest tube 4th–6th ICS safe triangle; ventilated/large leak → 24–28 Fr. [ACCP Baumann Chest 2001]

SCENARIO 5 — S5 · The refractory ventilated patient

ICU, hour 8. 75 kg PBW. The ventilator, the blood gas, and the blood pressure are all fighting you. Someone in the room has to make the calls.

Objectives: Attribute high Ppeak with acceptable Pplat to resistance · Set the vent: Vt 6–8 mL/kg, RR 10–14, VE ≤10, I:E ≥1:3, Pplat ≤30 · Diagnose hyperinflation shock with a disconnection trial · Distrust a low measured auto-PEEP (hidden auto-PEEP) · Trigger the ECMO referral on ELSO/EOLIA criteria; choose VV

> **Scene 1:** You inherit transport settings: A/C-VC, Vt 550, RR 20, flow 60 L/min, PEEP 5, FiO2 1.0.
> ✓ Best action: Huge peak-to-plateau gradient (>30) = RESISTANCE problem (bronchospasm), while Pplat reflects alveolar distension — the safety parameter

> **Scene 2:** Take the ventilator. Strategy: CONTROLLED HYPOVENTILATION / PERMISSIVE HYPERCAPNIA.
> ✓ Best action: Optimize to dossier bands: Vt 450–600 mL, RR 10–14, flow 60–100, set PEEP 0–5, VE ≤10, I:E ≥1:3, Pplat ≤30, ΔP ≤15
> ⌁ Discuss: Defend permissive hypercapnia to a consultant staring at pH 7.21. What are the REAL contraindications — and what does tolerating the CO2 buy this patient?
>
> - No RCTs — physiology plus series; the trade: accept PaCO2 50–90+ and pH >=7.20 to avoid the lethal cost of chasing normocapnia (hyperinflation, barotrauma, shock).
> - True contraindications: raised ICP, severe CV instability, sickle crisis. Bicarbonate below pH 7.20 is controversial/unproven.
> - Minute ventilation is the primary determinant of hyperinflation — the CO2 is the receipt, not the injury.

> **Scene 3:** Settings accepted — but MAP is 54 (BP 76/44) on norepinephrine. The nurse asks for a second pressor.
> ✓ Best action: DISCONNECT the ventilator 30–60 s (diagnostic + therapeutic), give volume, keep VE low
> ⌁ Discuss: The nurse asks why you are unplugging the ventilator from a crashing patient. Script your 20-second bedside explanation.
>
> - The chest is overfilled with trapped gas; every breath stacks more and squeezes venous return — disconnecting lets it out.
> - A transient BP rise during 30–60 s of disconnection CONFIRMS the mechanism, at zero cost.
> - Then: fluids, low minute ventilation, and only then more pressors — cause before number.

> **Scene 4:** End-expiratory hold reads auto-PEEP of only 5 cmH2O — yet Pplat is 28 and he remains hyperinflated on exam.
> ✓ Best action: HIDDEN auto-PEEP: with widespread airway closure the measured value underestimates true alveolar pressure — trust the Pplat

> **Scene 5:** Hour 14. Sevoflurane trial via AnaConDa: transient PIP improvement, now regressing.
> ✓ Best action: CALL CT SURGERY / ECMO TEAM NOW for VV ECMO — he meets criteria on every axis
> ⌁ Discuss: It is 2 AM and the nearest ECMO center is 90 minutes away. When should tonight's referral call have happened — and what does your region's pathway actually look like?
>
> - The worst prognostic factor is pre-ECMO arrest — the call belongs BEFORE the arrest, at criteria, not at futility.
> - Asthma is among the best ECMO indications (83.5–95% survival) — a low referral threshold is evidence-based, not soft.
> - Concretely: who answers your ECMO phone tonight? Mobile retrieval? What do they need from you? If you cannot answer, that is the homework.

> **Scene 6:** VV ECMO cannulated at hour 16. Within hours the ventilator is resting the lungs.
> ✓ Best action: Rapid de-escalation: reported series show PIP 38.2→25, driving pressure 29.5→16.8, FiO2 81→49 after ECMO — and the worst prognostic factor is pre-ECMO arrest, which you prevented

Debrief — key principles:

- High Ppeak + acceptable Pplat = resistance (peak–plateau gradient often >30); manage to Pplat ≤25–30 and ΔP ≤14–15. [Leatherman Chest 2015; Tobin NEJM 1994]
- Controlled hypoventilation: Vt 6–8 mL/kg PBW, RR 10–14, flow 60–100 square, set PEEP 0–5, VE ≤10 (often 6–8) — VE is the PRIMARY determinant of hyperinflation; I:E ≥1:3–1:4; exhalation needs ~3–5 tau; tolerate pH ≥7.20, PaCO2 50–90+ (contraindications: raised ICP, severe CV instability, sickle crisis). [Tuxen ARRD 1992; Corbridge AJRCCM 1995; Marini AJRCCM 2011]
- Hyperinflation shock (MAP 50–60): 30–60 s ventilator DISCONNECTION is diagnostic and therapeutic + fluids + lower VE. [Williams ARRD 1992; Caplan Front Physiol 2023]
- Hidden auto-PEEP: measured end-expiratory hold can read falsely low with airway closure (Ppeak 76/Pplat 28/auto-PEEP 5); trust Pplat; keep VEI <20 mL/kg (<1.4 L — ≥1.4 L had 65% complications). [Leatherman/Ravenscraft CCM 1996; Leatherman Chest 2015]
- ECMO triggers: ELSO (CO2 retention despite Pplat >30; P/F <100 on FiO2 >90%), Brodie/Bacchetta (pH <7.15 or Pplat >35–45), EOLIA-style (pH <7.25 + PaCO2 ≥60 >6 h; P/F <80 >6 h). VV preferred even if unstable; ECCO2R when mainly hypercapnic. Outcomes 83.5–95% survival; worst prognostic factor = pre-ECMO arrest. [ELSO / ISHLT-HFSA 2023; EOLIA; Yeo Crit Care 2017; Zakrajsek Chest 2023]

SCENARIO 6 — S6 · The clinic follow-up — after the storm

Six weeks later: the patient from S2–S5 survived ECMO and sits in your severe-asthma clinic. The next exacerbation is preventable — if this visit is done right.

Objectives: Separate difficult-to-treat from severe asthma · Order the severe-asthma workup that changes management · Phenotype for biologic eligibility (GINA + ACCP/CHEST) · Choose between overlapping biologic classes with the patient · Plan the 4-month trial, response review, and OCS-first de-escalation

> **Scene 1:** Six weeks post-ICU. On high-dose budesonide-formoterol MART since discharge; prednisone finished. ACT 16, one urgent-care wheeze visit last week. You watch him use the turbuhaler: he shakes it, fires it into a spacer, and inhales slowly — technique built for his old pMDI.
> ✓ Best action: DIFFICULT-TO-TREAT asthma for now — "severe" is a retrospective label applied only after technique, adherence, comorbidities and diagnosis are re-verified on optimized therapy
> ⌁ Discuss: "Difficult-to-treat does not mean a difficult patient." Which modifiable factors most often masquerade as severe asthma in your clinic — and which do we screen for worst?
>
> - Inhaler technique (device switches after hospitalization are a classic trap) and adherence top the list.
> - Comorbidities: chronic rhinosinusitis with nasal polyps, GERD, obesity, OSA, smoking/vaping.
> - Wrong diagnosis: inducible laryngeal obstruction, EGPA, ABPA — the "asthma" that is not asthma.
> - Payer reality: the severe label gates biologic access, so document every optimization step you take.

> **Scene 2:** Build today's severe-asthma workup. Select every test that will change management.
> ✓ Best action: Spirometry with bronchodilator response + Blood eosinophils — drawn now, while he is steroid-free + FeNO + Total IgE + aeroallergen sensitization (skin prick or specific IgE), plus a T2-comorbidity screen (nasal polyps, atopic dermatitis)
> ⌁ Discuss: Which single result on today's panel would most change your biologic choice — and which numbers will the payer demand?
>
> - Blood eosinophils: the anti-IL5/5Ra gate — and a value >1,500 flips the CHEST logic away from dupilumab.
> - FeNO: part of the T2-high definition, a strong dupilumab/tezepelumab predictor, and the CHEST switching biomarker (post-treatment FeNO >=25 advises dupilumab).
> - IgE + sensitization: an eligibility key for omalizumab that does NOT predict response magnitude.
> - Documented exacerbations and OCS courses — the currency of every prior authorization.

> **Scene 3:** Results (simulated): FEV1 61% predicted with bronchodilator response; blood eos 480/uL (steroid-free); FeNO 46 ppb; total IgE 220 IU/mL with dust-mite sensitization; nasal polyps on exam. Technique now correct, adherence verified — still ACT 15 with ongoing exacerbations.
> ✓ Best action: SEVERE T2-high asthma — uncontrolled despite verified, optimized high-dose therapy, with eos >=300 and FeNO >=25: biologic assessment is now appropriate
> ⌁ Discuss: His eosinophils were 80 during the ICU stay and 480 today. What does that swing teach about timing biomarkers — and what about the patient whose markers never "unmask"?
>
> - Steroids suppress eosinophils: GINA notes different cut-points may apply on OCS, and supervised weaning can unmask eligibility.
> - Serial measurements beat single draws; many payers accept the highest historical value.
> - For persistent T2-low presentations despite symptoms: tezepelumab retains efficacy without elevated T2 markers.

Scene 4: Profile: severe T2-high asthma; eos 480; FeNO 46; IgE 220 IU/mL + dust-mite sensitized; NASAL POLYPS; FEV1 61%; hospitalized (ECMO) this year; not currently OCS-dependent.
✓ Best action: Anti-IL5/5Ra or dupilumab, chosen WITH him — polyps (CRSwNP: dupilumab or mepolizumab), hospitalization, high eos/FeNO and FEV1 61% all point there; decide on comorbidity, route/frequency and payer criteria
⌁ Discuss: Four classes, one patient, zero head-to-head trials. In real life, how do you and the patient actually land on one?

- GINA's explicit tie-breakers: payer criteria, T2 comorbidities, response predictors, cost, dosing frequency, route/self-administration, patient preference.
- Logistics are legitimate medicine: q8-weekly benralizumab vs q2-weekly dupilumab is a lifestyle decision the patient owns.
- The polyps are a two-for-one: one drug, two diseases (CRSwNP: dupilumab or mepolizumab).
- Say the quiet part: prior-authorization criteria often decide before you do — document eos, FeNO, exacerbations, optimization.

Scene 5: Dupilumab is chosen with him (polyps + predictors + q2-week self-injection acceptable). Define success — and the plan if it fails.
✓ Best action: Trial at least 4 months; review at 3–4 months then every 3–6: ACT/ACQ-5, exacerbations & OCS courses, lung function, polyps, side-effects, satisfaction. Respond -> de-escalate OCS FIRST and never fully stop ICS; no response -> switch eligible classes
⌁ Discuss: He asks: "How long will I need this injection?" What do we know — and honestly not know — about stopping biologics?

- Re-evaluate every 3–6 months; de-escalate the OTHER therapies first: OCS, then add-on inhaled agents, then cautious ICS-LABA reduction after 3–6 months of stability.
- Discontinuation data are limited; relapse is common enough that most responders continue long-term.
- After long OCS exposure: adrenal-insufficiency surveillance — morning cortisol, plus stress-dose advice for up to 6 months after cessation.
- Practical honesty builds adherence: "we reassess every few months, and you help decide."

Debrief — key principles:

- Difficult-to-treat = uncontrolled despite medium/high-dose ICS + second controller; SEVERE is a retrospective subset label — uncontrolled despite verified adherence, optimized high-dose ICS-LABA, and managed contributory factors. [GINA SA Guide 2025]
- Optimize before biologics: technique, adherence, comorbidities, exposures; consider add-on LAMA/LTRA/low-dose azithromycin; low-dose OCS only as last resort; test/treat parasites before starting. [GINA SA Guide 2025]
- Eligibility gates: anti-IgE = sensitization + IgE/weight in dosing range (baseline IgE does not predict response); anti-IL5/5Ra = eos above local cut (>=150 or >=300); dupilumab = eos/FeNO or OCS-dependence, NOT if eos >1,500; tezepelumab = exacerbations, works even without elevated T2 markers. [GINA SA Guide 2025]
- CHEST comparative logic (conditional, very-low certainty): hospitalization, >=2 exacerbations/yr, or FEV1 <70% -> dupilumab over omalizumab; steroid-dependent -> anti-IL5/5Ra or dupilumab (OCS cuts ~47%/~28%), dupilumab over tezepelumab; eos >1,500 -> favor anti-IL5/5Ra; CRSwNP -> dupilumab or mepolizumab; post-treatment FeNO >=25 advises switching to dupilumab. [ACCP/CHEST Biologics 2025]
- Trial >=4 months; review at 3–4 months then every 3–6 (ACT/ACQ-5, exacerbations, OCS, lung function, comorbidities, satisfaction); good response -> OCS taper FIRST with adrenal surveillance; never completely stop ICS; non-response -> switch eligible classes. [GINA SA Guide 2025; ACCP/CHEST Biologics 2025]